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Sustainable Solutions for Global Challenges

Developing Virtual Reality for Museums in Saudi Arabia: A Technological Innovation for Sustainability and Enhanced Visitor Engagement

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Outline

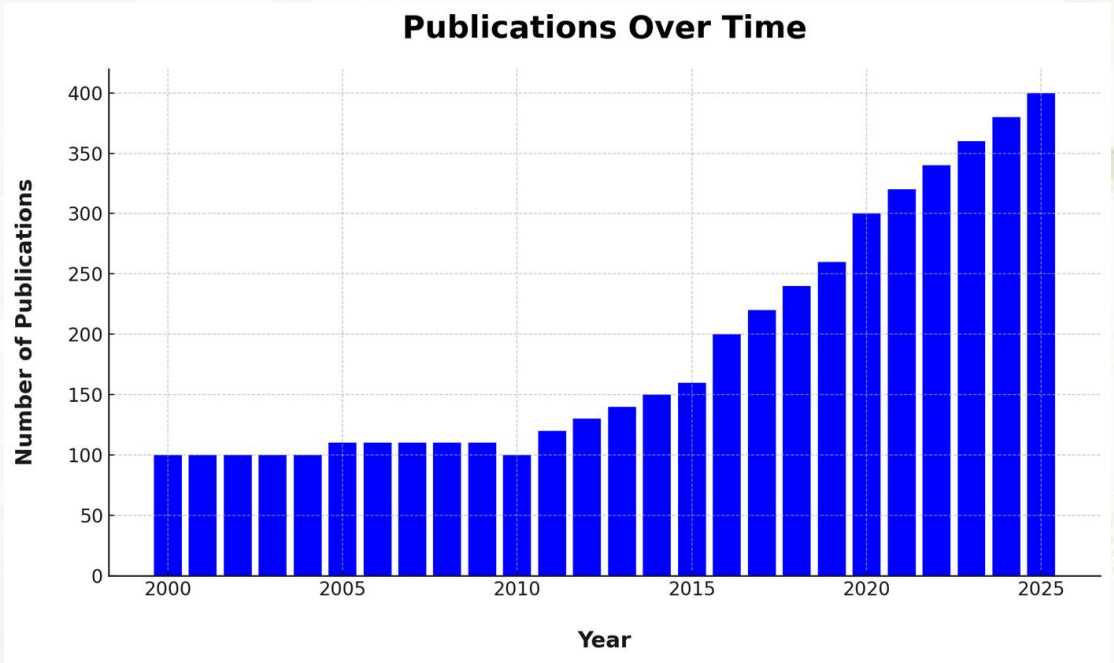
1. Introduction and Background
2. Problem Statement
3. Review of the State-of-the-Art
4. Research Methodology
5. Result & Discussion
6. Conclusion

Introduction & Background



Tayebat Museum - Visit Saudi Official Website

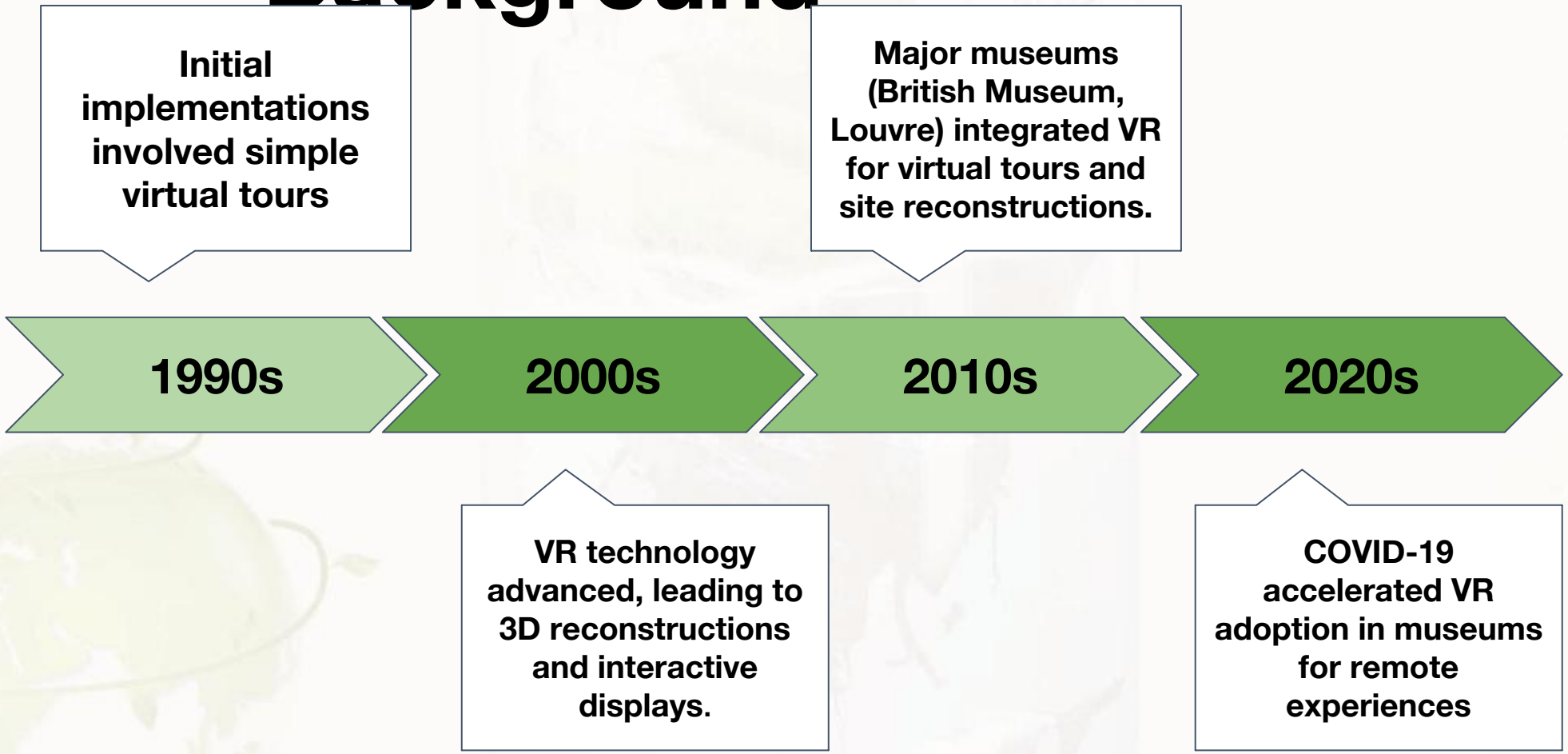
Integrating virtual reality into Al-Tayibat Museum



https://www.lens.org/

Interest in VR applications papers for museums has steadily grown since the early 2000s, reflecting the increasing recognition of VR's potential in cultural heritage.

Introduction & Background



Problem Statement

Research Question:
How Can the Implementation of Virtual Reality Technology Improve Visitor Engagement and Learning Experiences in Saudi Arabian Museums?

Research Aim:
To compare the virtual reality systems for museums and propose the best systems to use in Saudi Arabia museums.

Research Objectives:
Compare and evaluate global VR technologies for museums, identify gaps, and suggest recommendations based on the findings. And propose the best technologies to use in Saudi Arabian museums.

SIGNIFICANCE

Goal 1.1.2 & 1.1.3



SUSTAINABLE DEVELOPMENT GOALS

TARGET 4·7
EDUCATION FOR SUSTAINABLE DEVELOPMENT AND GLOBAL CITIZENSHIP

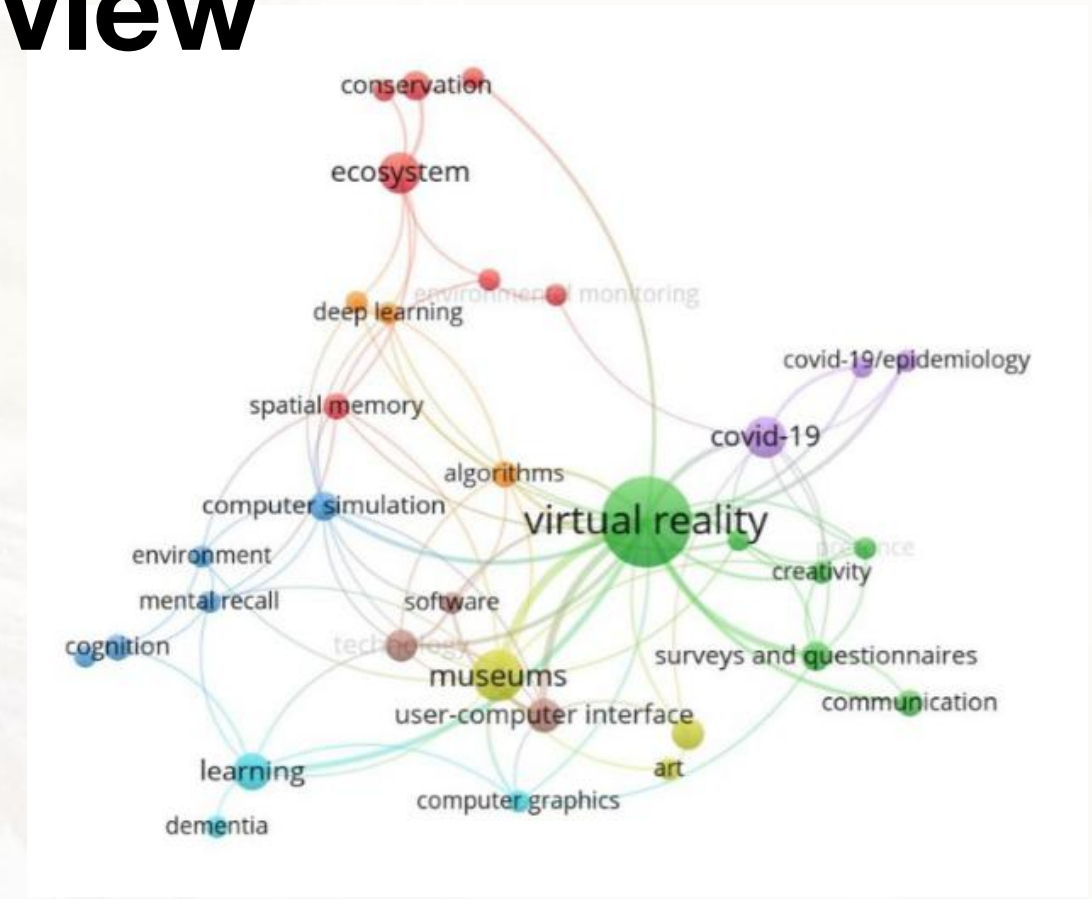
TARGET 9·5
ENHANCE RESEARCH AND UPGRADE INDUSTRIAL TECHNOLOGIES

TARGET 11·4
PROTECT THE WORLD'S CULTURAL AND NATURAL HERITAGE

State-of-the-Art Review

Table 2.1: Technologies Used in Museums

Technology	Description	Museums	Refs
The CrowdHeritage Platform	An ecosystem that improves cultural heritage metadata using crowdsourcing and AI, enabling users to annotate content while boosting public engagement.	Rijksmuseum in the Netherlands	[10, 11]
Computational Methods	The Louvre Museum in France uses VR technology to offer a virtual experience of the 'Mona Lisa.' Through advanced computational methods like 3D modeling and real-time rendering, visitors can explore the painting's history, restoration processes, and artistic techniques in a highly interactive and immersive format.	Louvre Museum in France	[19, 8]
Augmented Reality (AR)	Tate, a leading art institution in the UK, uses AR and VR to create immersive exhibits and expand access through online platforms.	Tate Museum in UK	[6, 1]
Web-based VR Platforms	Located in Balboa Park, this museum showcases art from around the world using virtual tours, like the SDMA 360, allowing remote visitors to explore the collection online.	San Diego Museum of Art in San Diego	[9, 2]
3D Photogrammetry	Enhances museum engagement by creating detailed 3D models of artifacts, enabling immersive exploration and bridging physical and digital experiences.	The British Museum in London, Louvre Museum in France	[17, 2]
The 2gether VR museum	A platform designed to create immersive environments, allowing visitors to explore artifacts and historical contexts. Uses interactive visual guides, animations, and emotions-on activities to foster emotional and intellectual connections with exhibits.	Platform remains in the experimental phase	[3, 15]



https://www.vosviewer.com/

The key themes are “virtual reality," "VR", and "museum.”

Literature Review



Table 2.2: Gaps Identified in VR Implementation in Museums

Gaps	Description	Reference
AI limitations	Automated tools struggle with domain-specific content, as AI models are often trained on datasets outside the cultural heritage domain, leading to inaccuracies.	[10, 11, 12]
Metadata consistency	There is still no streamlined process for metadata consistency and standardization across various cultural platforms.	[10, 19]
Skepticism and adoption	There remains some skepticism among art historians regarding the adoption of computational methods, partly due to their novelty and the perceived threat to traditional expertise.	[13, 17, 14]
Long-term impact of VR on visitor behavior	Limited studies on sustained visitor engagement or learning outcomes post-VR experiences.	[1, 5]
Cost-effectiveness of VR installations	Lack of detailed analysis on the long-term financial sustainability of VR in museums.	[7, 20, 15]
Integration of VR and traditional artifacts	Limited exploration of how to balance VR with physical artifacts in museum curation.	[1, 13]
User Familiarity with VR	Target audience may not be familiar with VR technology, affecting engagement.	[2, 7]
Image Quality Dependency	The success of digitization relies heavily on the quality of source images.	[17, 11, 7]
Complex Artifacts Handling	Difficulties in capturing complex shapes or delicate specimens accurately.	[17, 18]



Research Methodology

State-of-the-Art Review

Conduct a detailed review of existing VR technologies and algorithms to identify current capabilities and gaps.

Technology Analysis

Compare the technologies to identify the most suitable option for implementation in Saudi Arabia.

Software Requirements Specification (SRS)

A detailed document that outlines all system requirements, including user requirements, functional requirements, and non-functional requirements.

VR Development

Develop an VR prototype with algorithms optimized for sustainability and accessibility.

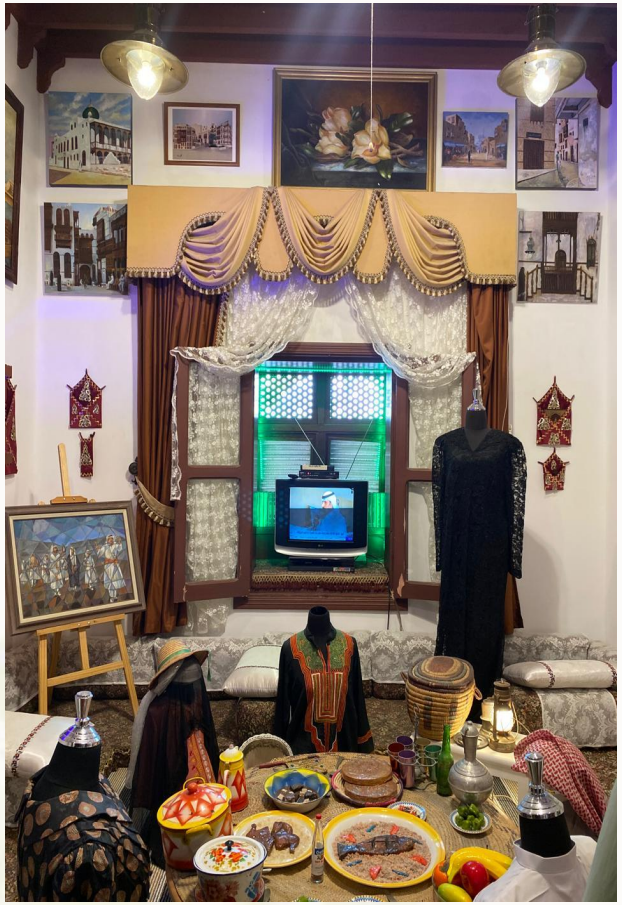
Potential Clients







Expected Results







Conclusion



- Virtual reality (VR) in Saudi Arabian museums has great potential to enhance visitor engagement and educational experiences, aligning with Saudi Vision 2030 and the UN's Sustainable Development Goals.
- We aim to identify the best VR solution for Saudi Arabian museums.
- Further research is needed to address accessibility, long-term visitor engagement, and the



Thank You

Any Questions?